

1(a)	kelvin, mole, ampere, candela <i>any two</i>	B1
1(b)	use of resistivity = RA/l and $V = IR$ (to give $\rho = VA/I l$)	C1
	units of V : (work done / charge) $\text{kg m}^2 \text{s}^{-2} (\text{A s})^{-1}$	C1
	units of resistivity: $(\text{kg m}^2 \text{s}^{-3} \text{A}^{-1} \text{A}^{-1} \text{m})$ $= \text{kg m}^3 \text{s}^{-3} \text{A}^{-2}$	A1
	or	
	use of $R = \rho L / A$ and $P = I^2 R$ (gives $\rho = PA / I^2 L$)	(C1)
	units of P : $\text{kg m}^2 \text{s}^{-3}$	(C1)
	units of resistivity: $(\text{kg m}^2 \text{s}^{-3} \times \text{m}^2) / (\text{A}^2 \times \text{m})$ $= \text{kg m}^3 \text{s}^{-3} \text{A}^{-2}$	(A1)
1(c)(i)	$\rho = (RA/l)$	C1
	$= (0.03 \times 1.5 \times 10^{-6}) / 2.5 \quad (= 1.8 \times 10^{-8})$	C1
	$= 18 \text{ n}\Omega \text{ m}$	A1
1(c)(ii)	1. precision is determined by the range in the measurements/values/readings/data/results	B1
	2. metre rule measures to $\pm 1 \text{ mm}$ and micrometer to $\pm 0.01 \text{ mm}$ (so there is less (percentage) uncertainty/random error)	B1